🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.8

Understand Absolute Value of Rational Numbers

Lesson 7-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the absolute value of a rational number — an integer, fraction, or decimal — as its distance from zero.

Language Objective: I can explain absolute value using the words distance, zero, opposite, and rational number.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Integers and Absolute Value | Integer | Positive | Negative | Absolute value | Opposite | Rational number

👆 *Tap any word to see what it means and a picture.*

1. The distance of an integer from zero is its ____________ ____________.

2. A whole number or its opposite, like −2, 0, or 5, is an ____________.

3. A number greater than zero is ____________.

4. A number less than zero is ____________.

5. The distance of a number from zero, always positive, is its ____________.

6 Two numbers the same distance from zero but on different sides, like 4 and −4, are ______.

7 Any number that can be written as a fraction is a ______ ______.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

How does absolute value help the rescue team compare the submarine and helicopter positions?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Integers and Absolute Value** — Integers are whole numbers and their opposites; absolute value is their distance from zero.
  Enteros y valor absoluto — Los enteros son números enteros y sus opuestos; el valor absoluto es su distancia desde cero.
- [ ] **Integer** — Whole numbers and their opposites, like -2, -1, 0, 1, 2.
  Número entero — Un número entero que puede ser positivo, negativo o cero.
- [ ] **Positive** — A number bigger than zero, to the right of zero on a number line.
  Positivo — Un número mayor que cero, a la derecha del cero en la recta numérica.
- [ ] **Negative** — A number smaller than zero, to the left of zero on a number line.
  Negativo — Un número menor que cero, a la izquierda del cero en la recta numérica.
- [ ] **Absolute value** — How far a number is from zero. It is never negative.
  Valor absoluto — Qué tan lejos está un número de cero. Nunca es negativo.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The submarine is at ____ meters, and its absolute value is ____ because ____. The helicopter is at ____ meters, and its absolute value is ____. Absolute value helps the rescue team because ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: -5 and 5 are the same distance from zero because ____.

Evidence: Student says both are 5 units from zero (same absolute value) but are opposites because of their signs.

Reasoning: This evidence connects the definition of absolute value to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The submarine is at ____ meters, and its absolute value is ____ because ____. The helicopter is at ____ meters, and its absolute value is ____. Absolute value helps the rescue team because ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Integers and Absolute Value
2. **Notes 2:** Integer
3. **Notes 3:** Positive
4. **Notes 4:** Negative
5. **Notes 5:** Absolute value
6. **Notes 6:** Opposite
7. **Notes 7:** Rational number
8. **Watch for this mistake:** *A common mistake in Integers and Absolute Value is judging size by the digits alone instead of position on the number line — students think -8 is 'more' than -3 because 8 > 3, when really -8 sits farther left (colder, deeper, or lower) and is actually the SMALLER integer. Absolute value measures distance from zero and is always positive or zero, but that distance does not decide which integer is greater — only left-to-right order on the number line does. Before you submit, ask: am I comparing distance from zero (absolute value) or position on the number line (greater/less than)?*
9. **Try It 1:** A. -5 degrees F (Ice Cave) *— On a number line, -5 is to the left of -2, so -5 is the smaller (colder) temperature.*
10. **Try It 2:** A. |-12| because 12 > 7 *— |-12| = 12 and |7| = 7. Since 12 > 7, the absolute value of -12 is greater.*